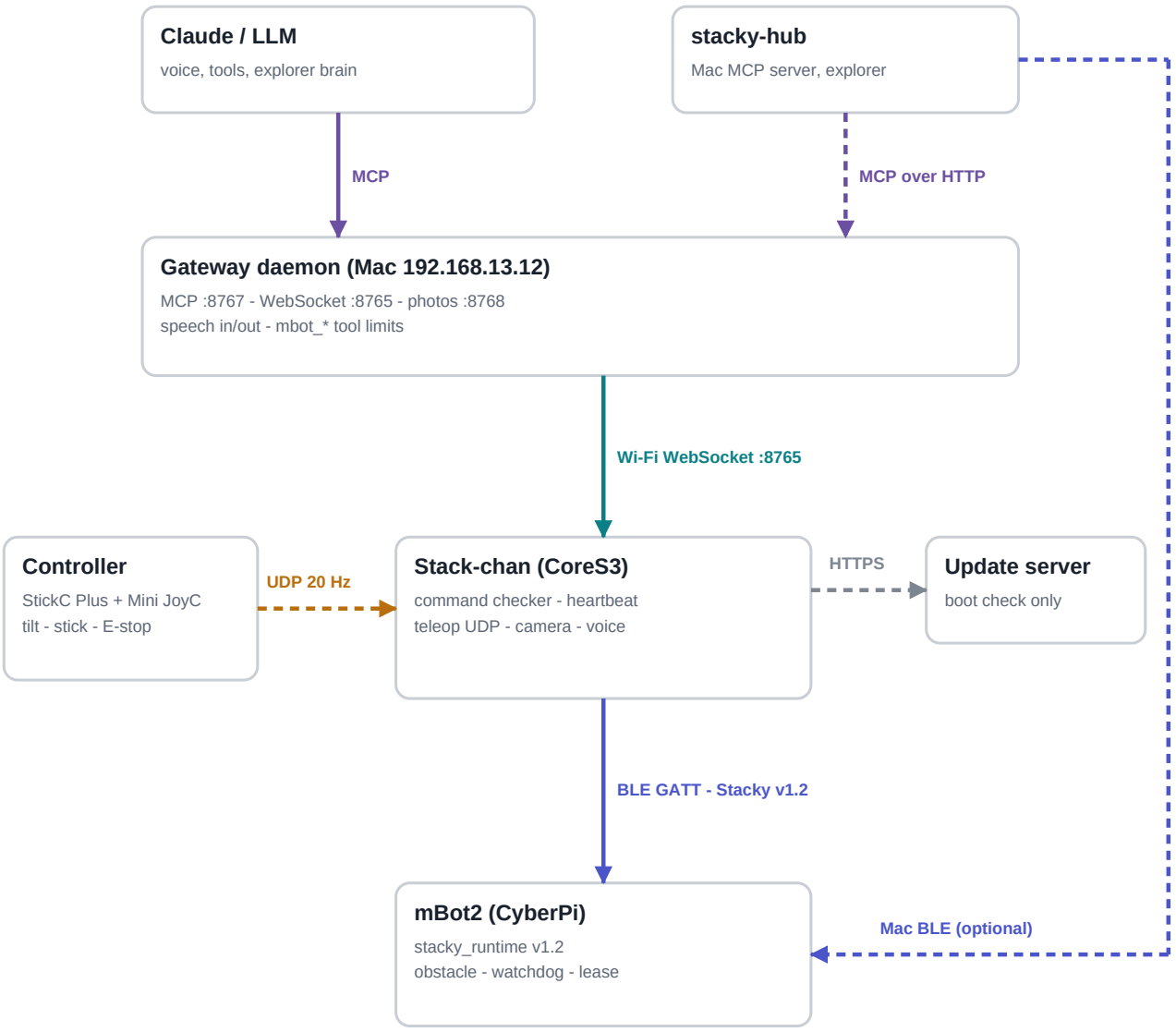
Stacky Link Map

Stack-chan, mBot2 and controller: architecture manual. 28 Sep 2026



Solid = verified on the hardware. Dashed = built, not yet tested on the hardware.

Links

Link	Transport	Carries	State
Claude -> gateway	MCP, localhost :8767	JSON-RPC tool calls	**Working**
stacky-hub -> gateway	MCP over HTTP	JSON-RPC tool calls	**Simulation only**
Gateway -> Stack-chan	Wi-Fi WebSocket :8765	MCP + audio + events	**Working**
Stack-chan -> mBot	BLE GATT, MTU 512	F3 frame + Stacky text	**Working (robot runs v1.1)**
Controller -> Stack-chan	Wi-Fi UDP :8790, 20 Hz	20-byte teleop packet	**Not tested yet**
Stack-chan -> update server	HTTPS, at boot	Version check	**Skipped after 1 failure**
Mac -> mBot	BLE or USB	F3 frame + Stacky text	**Working when Stack-chan is off**

Safety limits

Each limit is enforced in every layer listed. The mBot always has the last word.

Limit	Value	Enforced in	State
Wheel speed	max 60 rpm (about 20 cm/s)	Gateway, Stack-chan, mBot	**Active**
Timed move	max 5 s	Gateway, Stack-chan, mBot	**Active**
Step / turn	max 30 cm / 180 deg	Gateway, Stack-chan, mBot	**Active**
Arm S4	40-120 deg, home 90	Gateway, Stack-chan, mBot	**Active**
Gripper S3	45-120 deg, home 90	Gateway, Stack-chan, mBot	**Active**
Program	max 20 steps, 30 s	Gateway, Stack-chan, mBot	**Not tested on robot**
Obstacle	stop forward under 10 cm	mBot	**Active**
Watchdog	stop after 3 s of silence	mBot	**Active**
Buttons	B = stop + lock, A = unlock	mBot	**Active**
Drive lease	stop 300 ms after last drive	Stack-chan, mBot	**Needs mBot v1.2**
Dead-man	stop 300 ms after last packet	Stack-chan	**Not tested yet**
Tilt driving	wheels move only while A is held	Controller	**Not tested yet**
Limits in sync	all layers use the same values	tests/test_limits.py	**Active**

How to change it

To change	Edit	Then
A safety limit	docs/protocol.md, hub/stacky_hub/limits.py, mbot/micropython/stacky_runtime.py, firmware mbot_limits.h	Run pytest. test_limits.py fails if they differ.
An mBot command	stacky_runtime.py (parse_step, start_step), firmware mbot_command.cc, gateway mbot.py	Upload the runtime to the mBot. Flash Stack-chan.
The joystick packet	firmware teleop_packet.h, hub/stacky_hub/teleop.py, remote/joyc/src/teleop_packet.h	Update the golden vector in all three.
Controller buttons or modes	remote/joyc/src/main.cpp	Flash the StickC, then power-cycle it.
Teleop key or port	Kconfig STACKCHAN_MBOT_TELEOP_KEY / _PORT, remote/joyc/include/config.h	Flash Stack-chan and the StickC.
Stack-chan Wi-Fi	Hotspot Xiaozhi-3679, page 192.168.4.1	Keep the Mac on the same network.
Gateway address	gateway_config_set tool	Stack-chan reconnects by itself.

Stack-chan memory

Internal RAM: 341 KB. Built-in use: 165 KB. Free before Bluetooth: 55-65 KB. Free after Bluetooth: 25-35 KB. **Lowest seen: 11.7 KB.** Low memory breaks Bluetooth, TLS or Wi-Fi.

Next fix	Saves	Risk
Add a memory report tool	measure	none
Turn off face detection in the mBot build	about 24 KB	low
Move the audio encoder stack to PSRAM	about 24 KB	medium

Open items

1. Upload stacky_runtime.py v1.2 to the mBot (needed for any drive command).
2. Flash Stack-chan with the teleop listener.
3. Check the controller tilt directions, then do a first capped drive.
4. Check the head direction, then run a first supervised exploration.